

MLPERF STORAGE V3.0 · CLOSED SUBMISSION

AIStore on Google Cloud Platform

Three-node GCP shared object storage with eight benchmark clients

aistore-3n-gcp-z3-200g-8client-n2d96

SUBMITTER

NVIDIA

BENCHMARK

MLPerf Storage v3.0
mlpstorage 3.0.46

SOLUTION TYPE

Open-source shared object storage

POWER REQUIREMENTS

Provider managed; cloud exemption applies

SUBMISSION DATE

July 24, 2026

DIVISION

CLOSED

DEPLOYMENT

Cloud · Kubernetes

SUBMITTED WORKLOAD

UNet3D training

8 benchmark clients

n2d-standard-96



3-node AIStore cluster

3 proxies · 3 targets · 36 target NVMe devices

System topology and measured data path

Clients access AIStore exclusively through its S3-compatible API. A proxy admits each request and redirects the object transfer to the responsible target.

8 × n2d-standard-96 benchmark clients

Open MPI · DLIO · s3dlcio · equal rank placement per participating host

↓ S3 over HTTP through three supplied proxy endpoints selected round-robin

3 × AIStore proxy pods

Request admission, cluster map, placement, and HTTP redirect

↓ direct client-to-target redirected object transfer

3 × AIStore target pods on z3-highmem-176-standardlssd

36 local NVMe devices · one native distributed object namespace

Architecture reference

For AIStore proxy, target, routing, and deployment details, see the [AIStore architecture overview](#).

Cloud infrastructure inventory

8

BENCHMARK CLIENTS

3

AISTORE STORAGE NODES

36

TARGET NVME DEVICES

Component	Benchmark clients	AIStore storage nodes
Quantity / role	8 external MPI hosts	3 Kubernetes nodes; one proxy and one target per node
Cloud shape	n2d-standard-96	z3-highmem-176-standardlssd
Compute / network	2 × AMD EPYC 7B13; 48 captured cores; 384 GB captured memory; 100 Gb/s data networking	2 × Intel Xeon Platinum 8481C; 88 captured cores; 1,408 GB captured memory; 200 Gb/s data networking
Local media	Present but not used for MLPerf data	12 × 3 TB TLC NVME per target; 36 devices total
Platform	Ubuntu 24.04	Google Container-Optimized OS kernel 6.12.85+

Client-role clarification

Client persistent disks were used for the operating system and result logs only. Client-local storage was not used for benchmark data. All submitted objects were stored by the AIStore targets.

Storage capacity

- 108.00 TB aggregate target-drive nameplate capacity.
- Native AIStore bucket on local target NVMe.
- Mirroring and erasure coding disabled.

Cloud-managed facilities

- Physical switching, racks, cooling, and PSUs are provider managed.
- The MLPerf cloud-deployment power and rack-unit exemption applies.

Reproducible software stack

Layer	Version / configuration used	Reference
AIStore	v4.8, commit 42242b334fd57e993919f04acf3c2011cf2db675	source
MLPerf Storage	mlpstorage 3.0.46; captured code image(s) 3a9a7bf3c80740e7e0a2f70688152c9c f9327d87f7078a8b2e0251370decd883	1a236dc3e00b0578a8c03d7ed81b3f54d0329637
I/O stack	DLIO with s3dlcio; path-style S3 over HTTP	DLIO
Client platform	Ubuntu 24.04	GCP n2d-standard-96 clients / z3-highmem-176-standardlssd storage
Storage platform	Google Container-Optimized OS kernel 6.12.85+	GCP n2d-standard-96 clients / z3-highmem-176-standardlssd storage

Object path

- Native AIStore bucket through the S3-compatible API.
- Path-style S3 and standard AIStore HTTP redirects.
- Recorded endpoint selection: three supplied proxy endpoints selected round-robin.

MPI placement

- Equal ranks per participating client.
- `--bind-to none --map-by node`.
- Client-local NVMe outside the data path.

Captured non-default S3 and TCP settings

`S3DLIO_FOLLOW_REDIRECTS=1`
`S3DLIO_RANGE_THRESHOLD_MB=1024`
`S3DLIO_MULTIPART_THRESHOLD_MB=4096`
`S3DLIO_OPERATION_TIMEOUT_SECS=300`
`S3DLIO_CONNECT_TIMEOUT_SECS=60`
`S3DLIO_MAX_RETRY_ATTEMPTS=10`
`S3DLIO_POOL_MAX_IDLE_PER_HOST=64`
`AWS_MAX_ATTEMPTS=10`
`AWS_RETRY_MODE=standard`
`net.ipv4.tcp_tw_reuse=1`
`net.ipv4.tcp_retries2=15`

The matching YAML is the authoritative per-host hardware, operating-system, network, environment, sysctl, drive, and capability inventory. Secret values are redacted.

CLOSED workload configuration

This page records the submitted workload parameters and a compact snapshot of the validated result artifacts.

Parameter	Submitted configuration
Model / accelerator	UNet3D / B200
Scale	8 ranks; 1 per client
Dataset	1 12,501 NPZ training objects; approximately 15 TiB
Reader	batch size 7; reader.read_threads=4
Invocation	5 epochs; one warm-up plus five measured runs

Validated result snapshot

Workload	Mean AU	Samples/s	Training I/O
UNet3D	97.88%	338.04	46.15 GiB/s

Interpretation

The mean AU, samples/s, and I/O values are the aggregate values reported by the validated MLPerf artifacts. The corresponding packaged result files remain authoritative.

Primary sources and submission scope

These links connect the tested configuration to product specifications, source code, and protocol behavior.

NVIDIA AIStore documentation<https://docs.nvidia.com/aistore>**AIStore architecture and deployment portability**<https://docs.nvidia.com/aistore/overview>**AIStore S3 compatibility**<https://docs.nvidia.com/aistore/s3compat>**AIStore open-source repository**<https://github.com/NVIDIA/aistore>**Cloud instance specification**<https://cloud.google.com/compute/docs/machine-resource>**MLCommons Storage benchmark and rules**<https://github.com/mlcommons/storage>**Exact AIStore source revision**[42242b334fd57e993919f04acf3c2011cf2db675](https://github.com/NVIDIA/aistore/commit/42242b334fd57e993919f04acf3c2011cf2db675)**Document scope**

This PDF describes only aistore-3n-gcp-z3-200g-8client-n2d96 and the result artifacts stored under that system ID. Other client counts, workload configurations, software revisions, and clouds have separate same-name YAML/PDF descriptions.

Machine-readable companion

The same-name YAML is the authoritative structured system inventory.

Validated artifacts

Logs, metadata, resolved configurations, summaries, code-image markers, and aggregate results remain in the matching results directory.

Important context and disclosures

- The validated GCP UNet3D point achieved 46.15 GiB/s, 66.1% of the aggregate nominal 3×200 Gb/s storage-interface reference. The submitted configuration kept accelerator utilization (AU) compliant and performance stable across all six runs; it was not tuned to saturate the aggregate interfaces, and different client or rank settings may produce different throughput.
- OCI, AWS, and GCP three-node UNet3D results are not like-for-like cloud comparisons. Captured storage/client networking is 100/100 Gb/s on OCI, 150/300 Gb/s on AWS, and 200/100 Gb/s on GCP; shapes, client counts, datasets, rank counts, and workload tuning also differ.
- This submission used HTTP to keep the benchmark path simple and consistent. AIStore [supports HTTPS/TLS](#). With HTTPS, AIStore disables the file-to-socket sendfile fast path (encrypted responses use buffered transfer, and TLS encryption and handshakes add CPU work), so HTTPS performance may differ.
- NVIDIA AIStore is not a managed service offering. It is open source, and its [source code](#), [deployment guidance](#), and [deployment recipes](#) are publicly available. These are deployment-specific measurements, not performance guarantees or competitive claims.

Supporting documentation

The following documents provide disclosure, scaling, and deployment context applicable to this NVIDIA AIStore system.

[AIStore Disclosure](#)

[AIStore Scaling and Cloud Portability](#)